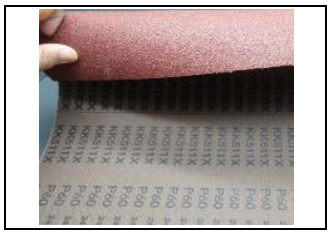
STANDARD OPERATING PROCEDURE OF DIN ABRASION VF-C01

Lab /SOP/ 15



DIN Abrasion machine
Drilling die, 16.0 +/-0.2 mm in diameter
Electric drill mounted in a drill stand.
Cyanoacrylate (super-glue)
Aluminum disc, diameter of 16.0+/-0.2 mm,
and minimum thickness of 6mm.
Balance, capable of weighing to the nearest 0.0001 grams.



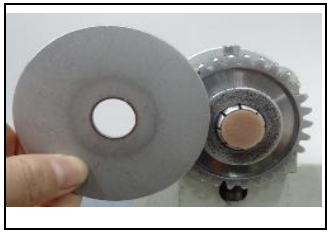
Corundum abrasive cloth A of grain size 60
bonded to cotton twill



Standard elastomer A, meeting the requirements
of DIN 4649.
After every ten (10) test runs, the "standard elastomer"
test sample must be evaluated to ensure that the
equipment is properly calibrated.
standard elastomer" produces an average weight
loss of 200+/-20 mg



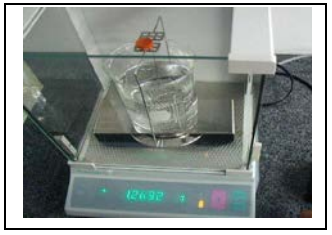
One piece of the "standard" elastomer is to be measured
three (3) times, and averaged.
Using the drilling die, cut three (3) test samples
have a diameter of 16.0+/- 0.2 mm
Determine the specific gravity (SG) of the test
material.



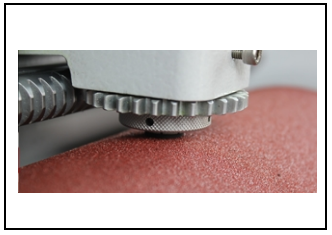
Pre-buff the test sample before the test by
allowing it to rotate and traverse at least
half of the length of the drum.
Remove any burrs, and then brush any dust
or loose particles from the sample with a soft
bristle brush.
Weigh the test sample to the nearest 0.001 gram (W1).



Insert the test sample in the sample holder of
the DIN abrader
the test sample protruding 2.0 +/-0.2 mm
beyond the end of the retaining rim.
Load the sample holder with the total weight
of the sample holder to 1 kg.
Lower the sample holder to the starting position
turn on the machine and vacuum



At the completion of testing, remove the test sample
from the holder, remove any burrs with your fingers,
and brush any dust or loose particles from the sample
with a soft bristle brush.
Re-weigh the test sample to the nearest 0.001 gram (W2).



DIN Abrasion (volume loss in mm3) =

$$\frac{(W1-W2) \times 200mg \times 1000}{SG \times S_A}$$

SA= Actual reading of the standard elastomer
The average of the three test samples, to the nearest
one (1) mm3, as final test result.

Prepared by	Approved by	REVISION		DATE	